Day 150: Exponential Smoothing from Scratch: Simple, Holt and Holt-Winters, against ARIMA

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 22, Day 4 | Complete study guide

Watch: Exponential Smoothing: Trends and Seasonality (Kevin Sheppard) | Smoothing 6: Winter's exponential smoothing (Galit Shmueli)

Script: `day150_exponential_smoothing.py` · every number in this guide is copied by program from `day150_run_output.txt`, the real output of that script.

Exponential smoothing keeps running estimates of a series' level, trend and season, and nudges each one towards every new month. Today builds simple smoothing, Holt and Holt-Winters from scratch, fits them with Day 149's solver and checks them against statsmodels: same one-step forecasts (1.1e-13), same fitted weights. The check also found two forecast steps where statsmodels 0.15.0 differs from the textbook formula, and a self-consistency test shows ours is the consistent one. On CO2, Holt-Winters chosen by AIC forecast 2000–2001 with a mean error of 0.329 ppm, better than Day 149's ARIMA (0.412). Over 97 rolling origins the two are nearly tied, but Holt-Winters' intervals were too narrow (coverage as low as 0.835).

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

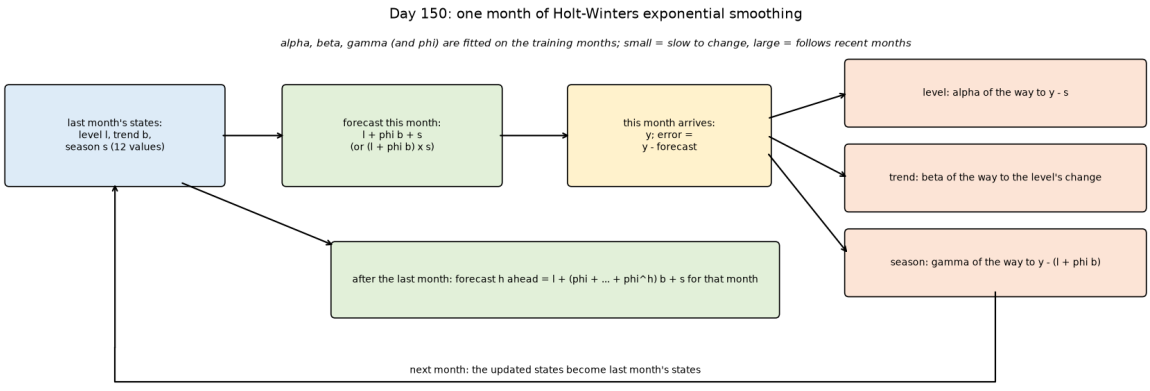
Exponential smoothing forecasts with running estimates. After each month, the estimated **level** moves a fraction alpha of the way towards what the month showed; the **trend** moves a fraction beta towards the level's latest change; each month's **seasonal** value moves a fraction gamma towards how far that month sat from the level. Small fractions mean slow, steady estimates; large ones follow recent months closely.

Day 149's ARIMA modelled the differenced series' memory. Smoothing models the level, trend and season directly, and section 1 shows that the two meet in one special case.

REAL-WORLD SCENARIO

Estimating your usual commute time: after each trip you shift your estimate a little towards today's time. A large shift reacts fast to roadworks but also to one-off jams; a small shift is steady but slow to notice that the roadworks have started. If Mondays are always slower, you also keep a separate adjustment for each weekday: that is the seasonal part.

SYSTEM DIAGRAM



One month of Holt-Winters: forecast, compare, update the three estimates.

Today's part for the week's build

`day150_exercise.py` is today's parts (the smoothing equations for every variant, the starting values, the least-squares fit from four starts, the forecasts and the interval formula), checked by hand: three steps of simple smoothing, a Holt line followed exactly, the starting values of a tiny seasonal series, the fitted alpha against a brute-force search, and the interval for simple smoothing.

ACTUAL OUTPUT

```
SES, alpha 0.5, start 10, data 10, 12, 11: forecasts 10, 10, 0.5 x 12 + 0.5 x 10 = 11; final level
    0.5 x 11 + 0.5 x 11 = 11; ets_filter gives [10.0, 10.0, 11.0], level 11.0
Holt, alpha 1 and beta 1, on the line 1, 2, 3, 4 (start level 0, trend 1): forecasts [1.0, 2.0,
    3.0, 4.0], then [5.0, 6.0, 7.0]
ets_init on 11, 9, 13, 11 with a season of 2: level (11 + 9) / 2 = 10, trend (12 - 10) / 2 = 1,
    season 1 and -1; gives [10.0, 1.0], [1.0, -1.0]
SES alpha on 200 simulated values: brute force over 999 values 0.556, fit_ets 0.556
interval for SES, alpha 0.5, error variance 1: sd 3 steps ahead = sqrt(1 + 2 x 0.25) = 1.2247;
    ets_sd gives 1.2247
```

Data: Day 147's two series. CO2 is trained up to 1999-12 and tested on 2000–2001, as on Days 147 and 149; the orders are used on a log scale, where their season adds. The smoothing weights and the choice of variant come from the training months only. Starting states come from the first two years (a simple rule, fixed rather than fitted).

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. Simple exponential smoothing

DEFINITION — start here, no prior knowledge assumed

Simple exponential smoothing (SES) keeps only a level. Each month: new level = alpha × this month + (1 – alpha) × old level. Unrolled, this is a weighted average of all past months in which the weight of the month k steps back is alpha $(1 - \text{alpha})^k$: geometric decay. The forecast for every future month is the last level, a flat line.

SES is also a special case of Day 149's ARIMA: ARIMA(0,1,1) with theta = alpha – 1 produces exactly the same one-step errors.

REAL-WORLD SCENARIO

A thermostat that reads the room every minute and averages, giving most weight to the latest readings, so a door opened briefly barely moves it, while a real change of weather does, gradually.

MATHEMATICS

$$l_t = \alpha\, y_t + (1-\alpha)\, l_{t-1} = \alpha\, y_t + \alpha(1-\alpha)\, y_{t-1} + \alpha(1-\alpha)^2\, y_{t-2} + \ldots; \quad forecast = l_t$$

$$error\ e_t = y_t - l_{t-1}: \quad y_t - y_{t-1} = e_t + (\alpha - 1)\, e_{t-1} \quad (ARIMA(0,1,1),\ \theta = \alpha - 1)$$

ACTUAL OUTPUT

```
the level after each month: level = alpha x this month + (1 - alpha) x the old level; unrolled, month k back gets
weight alpha (1 - alpha)^k
  alpha 0.2: weights of this month and the 3 before [0.2, 0.16, 0.128, 0.102]; last 12 months together 0.931
  alpha 0.5: weights of this month and the 3 before [0.5, 0.25, 0.125, 0.062]; last 12 months together 1.000
  alpha 0.8: weights of this month and the 3 before [0.8, 0.16, 0.032, 0.006]; last 12 months together 1.000
a wandering level plus noise (300 values): fitted alpha 0.4273; SES's one-step errors against ARIMA(0,1,1) with theta =
    alpha - 1 = -0.5727: largest difference 4.4e-15
so SES is a special case of Day 149's model: both say the series is a random walk seen through noise
Actual output from this lesson's run.
```

What it means

With alpha 0.2 the last 12 months hold 0.931 of the weight; with alpha 0.8 the latest month alone holds 0.8. On a simulated wandering level the fitted alpha was 0.4273, and SES's errors equal ARIMA(0,1,1)'s with theta -0.5727 to 4.4e-15: the same model written two ways.

WHY CHOOSE THIS?

One weight, one recursion, and a forecast anyone can explain.

WHY NOT THIS? (TRADEOFFS)

A flat forecast: no trend, no season. On CO2 it can only repeat the last month (section 3 shows alpha going to 1).

2. Holt and Holt-Winters, checked against statsmodels

DEFINITION — start here, no prior knowledge assumed

Holt adds a trend: the forecast becomes a straight line. A **damped** trend multiplies the trend by phi < 1 each step, so long-range forecasts level off. **Holt-Winters** adds one seasonal value per month, either added to the level (additive) or multiplying it (multiplicative, for a season that grows with the level).

The weights are fitted by least squares on the one-step errors, using Day 149's Levenberg-Marquardt solver. Each weight is kept between 0 and 1 by fitting an unconstrained number u and using $1 / (1 + e^{-u})$.

REAL-WORLD SCENARIO

A shop's sales: a level (typical month), a trend (growing 2% a year) and a season (December is double). Each is updated as every month's figures come in.

MATHEMATICS

$$forecast:\ y\text{-}hat = l + \phi\, b + s_{month} \quad (multiplicative: (l + \phi\, b) \times s_{month})$$

$$l \leftarrow \alpha(y - s) + (1-\alpha)(l + \phi\, b); \quad b \leftarrow \beta(l_{new} - l) + (1-\beta)\, \phi\, b$$

$$s \leftarrow \gamma(y - l - \phi\, b) + (1-\gamma)\, s \quad (statsmodels'\ form:\ the\ old\ level\ and\ trend)$$

$$h\ steps\ ahead:\ l + (\phi + \phi^2 + \ldots + \phi^h)\, b + the\ latest\ s\ for\ that\ month$$

IMPLEMENTATION

```
def ets_filter(y, alpha, beta, gamma, phi, l0, b0, s0, trend, season, m=M):
    l, b, s = l0, b0, np.array(s0, float).copy()
    fit = np.empty(len(y)); hist = np.empty(len(y))
    for t, yt in enumerate(y):
        st = s[t % m]
        base = l + phi * b if trend else l
        if season == "mul":
            fit[t] = base * st; ln = alpha * yt / st + (1 - alpha) * base
            s[t % m] = gamma * yt / base + (1 - gamma) * st
        elif season == "add":
            fit[t] = base + st; ln = alpha * (yt - st) + (1 - alpha) * base
            s[t % m] = gamma * (yt - base) + (1 - gamma) * st
        else:
            fit[t] = base; ln = alpha * yt + (1 - alpha) * base
        if trend:
            b = beta * (ln - l) + (1 - beta) * phi * b
        l = ln; hist[t] = s[t % m]
    return fit, l, b, s, hist
```

IMPLEMENTATION

```
def fit_ets(y, trend, season, damped, m=M, starts=STARTS):
    l0, b0, s0 = ets_init(y, m, season or "add")
    use = [True, trend is not None, season is not None, damped]
    def unpack(u):
        p = iter(1 / (1 + np.exp(-u)))
        a, b, g, f = (next(p) if k else d for k, d in zip(use, (0.0, 0.0, 0.0, 1.0)))
        return a, b, g, (0.8 + 0.18 * f if damped else 1.0)
    def res(u):
        return y - ets_filter(y, *unpack(u), l0, b0, s0, trend, season, m)[0]
    best = None
    for st in starts:
        p0 = np.array([v for v, k in zip(st, use) if k]); u, ssr = d149.levenberg_marquardt(res,
            np.log(p0 / (1 - p0)))
        if best is None or ssr < best[1]:
            best = (u, ssr)
    a, b, g, f = unpack(best[0]); _, l, bb, s, hist = ets_filter(y, a, b, g, f, l0, b0, s0, trend,
        season, m)
    return dict(alpha=a, beta=b, gamma=g, phi=f, sse=best[1], n=len(y), k=sum(use), level=l,
        slope=bb, season=s, hist=hist,
                trend_kind=trend, season_kind=season, init=(l0, b0, s0))
```

ACTUAL OUTPUT

```
  fixed weights, add          one-step forecasts: largest difference 1.1e-13
  fixed weights, mul          one-step forecasts: largest difference 0.0e+00
  fixed weights, add, damped  one-step forecasts: largest difference 1.1e-13
  fitted, add: ours alpha 0.4646 beta 0.0116 gamma 0.3066, sum of squares 60.7544; statsmodels 0.4646 0.0116 0.3066, 60.7544
  fitted, mul: ours alpha 0.4630 beta 0.0116 gamma 0.2949, sum of squares 60.5428; statsmodels 0.4630 0.0116 0.2949, 60.5428
  forecasts with the same weights: equal at 22 of 24 steps; steps [12, 24] differ by -0.0468
  at step 12 statsmodels used -0.3778 as the seasonal value: the value before the last update (-0.3778), not the latest
      (-0.4246)
  self-check: feed the 12 forecasts back in as data (every error 0); each one-step forecast should equal the forecast:
    ours 1.1e-13; statsmodels' 4.7e-02 (statsmodels 0.15.0)
Actual output from this lesson's run.
```

What it means

With the same weights and starting values, our one-step forecasts equal statsmodels' (1.1e-13 additive, 0.0e+00 multiplicative), and our fit reaches the same weights and the same sum of squares (60.7544, 60.5428).

The forecasts agree at 22 of 24 steps. At steps 12 and 24 statsmodels used the seasonal value from before the last update (-0.3778) rather than the latest (-0.4246). Which is right? A self-check: pretend the next 12 months arrive exactly as forecast, so every error is 0; each one-step forecast should then equal the forecast. Ours does (1.1e-13); statsmodels' does not (4.7e-02). Whether this is intended in statsmodels was not determined here.

WHY CHOOSE THIS?

Every variant is the same short recursion with parts switched on or off, and fits in a fraction of a second.

WHY NOT THIS? (TRADEOFFS)

The starting states are a simple rule here; statsmodels can also fit them, which would change the weights slightly.

3. Choosing the variant for CO2

DEFINITION — start here, no prior knowledge assumed

Seven variants, each fitted on the training months: SES, Holt, damped Holt, and Holt-Winters additive and multiplicative, each with and without damping. AIC compares them (as on Day 149: a smaller error variance, minus 2 for each weight), and the smallest AIC is the choice. Every fit starts from four different points and keeps the best, because the fit can get stuck.

REAL-WORLD SCENARIO

Trying on shoes: several styles, one score that weighs comfort against weight, and the choice is made in the shop, not after the hike.

ACTUAL OUTPUT

```
                      variant   alpha    beta   gamma    phi  sum of sq.       AIC
                          SES  1.0000  0.0000  0.0000  1.000    709.0739    175.37
                         Holt  1.0000  1.0000  0.0000  1.000    415.0547    -91.48
                Holt, damped  1.0000  1.0000  0.0000  0.800    360.6563   -160.00
       Holt-Winters, additive  0.4646  0.0116  0.3066  1.000     60.7544  -1054.10
 Holt-Winters, multiplicative  0.4630  0.0116  0.2949  1.000     60.5428  -1055.86
          HW additive, damped  0.4811  0.0492  0.2916  0.980     64.7418  -1020.19
    HW multiplicative, damped  0.4776  0.0508  0.2807  0.980     64.5628  -1021.58
(6.5 s) AIC's choice: Holt-Winters, multiplicative
first draft, three starts: Holt stopped at a sum of squares of 704.7799 (alpha 1.0000, beta 0.0000);
the fourth start (alpha 0.9, beta 0.9) finds 415.0547 (alpha 1.0000, beta 1.0000): a local minimum, and several starts are
   no guarantee
Actual output from this lesson's run.
```

What it means

AIC chose **Holt-Winters, multiplicative** (-1055.86), just ahead of additive (-1054.10); damping made both worse (-1021.58). Without a season the fits are poor: SES pushed alpha to 1.0000, which simply repeats the last month.

A first-draft failure, kept on purpose: with three starting points, Holt stopped at a sum of squares of 704.7799 (alpha 1, beta 0). A fourth start (alpha 0.9, beta 0.9) found 415.0547 at alpha 1, beta 1. The fit has local minima, and several starts are no guarantee; every fit now uses four.

WHY CHOOSE THIS?

AIC picks the variant on the training months by a rule, keeping the test honest.

WHY NOT THIS? (TRADEOFFS)

Close AIC values (here 1.76 apart) mean either choice is defensible; the test may favour the other one, as section 4 shows.

4. Forecasting 2000–2001

DEFINITION — start here, no prior knowledge assumed

The same 24 test months as Days 147 and 149, with every earlier forecast in one table. For the additive model there is also an interval: each future error carries forward with weight c_j, so the variance grows with the horizon.

REAL-WORLD SCENARIO

Several forecasters predicting the same two years, scored on the same months: a fair contest only because nobody saw the answers first.

MATHEMATICS

additive, undamped: $var(h \text{ steps}) = \sigma^2 (1 + c_1^2 + \ldots + c_{h-1}^2)$, $c_j = \alpha(1 + j\beta) + \gamma$ *[j a multiple of 12]*

IMPLEMENTATION

```
def ets_forecast(F, h, m=M):
    steps = np.cumsum(F["phi"] ** np.arange(1, h + 1)) if F["trend_kind"] else np.zeros(h)
    base = F["level"] + steps * F["slope"]
    s = F["season"][(F["n"] + np.arange(h)) % m]
    return base + s if F["season_kind"] == "add" else base * s if F["season_kind"] == "mul" else
        base
def ets_sd(F, h, m=M):
    j = np.arange(1, h)
    c = F["alpha"] * (1 + F["beta"] * j) + F["gamma"] * (j % m == 0)
    return np.sqrt(F["sse"] / F["n"] * np.r_[1.0, 1 + np.cumsum(c ** 2)])
```

ACTUAL OUTPUT

```
2000-01 to 2001-12, 24 months             MAE (ppm)   bias
  same month last year (Day 147)           1.882  -1.882
  airline ARIMA, CSS (Day 149)             0.412  +0.405
  airline ARIMA, MLE (Day 149)             0.359  +0.359
  SES                                      2.420  -2.207
  Holt                                    13.418 +13.418
  Holt, damped                             2.178  +1.963
  Holt-Winters, additive                   0.290  +0.273
  Holt-Winters, multiplicative <- AIC      0.329  +0.314
  HW additive, damped                      0.523  -0.423
  HW multiplicative, damped                0.489  -0.361
Holt-Winters additive 95% interval: half-width 0.68 one month ahead, 1.92 at 24; 24 of 24 months inside (the airline
    model's: 2.40 at 24)
AIC's choice and the additive version are close on the training data; on these 24 test months the additive one did
better -- picking it now would be choosing on the test, so the comparison stands as AIC chose
Actual output from this lesson's run.
```

What it means

AIC's choice forecast with a mean error of 0.329 ppm, better than the airline ARIMA (0.412 CSS, 0.359 MLE) and far better than same month last year (1.882). The additive version did even better (0.290), but choosing it now would be choosing on the test. The variants without a season fail: SES only repeats the last month (2.420), and Holt, with its better-fitting beta of 1, extends the last month-to-month change, which in a seasonal series is mostly season: 13.418. A better fit to the training months gave a far worse forecast. The additive interval held all 24 of 24 months and is narrower than the airline model's (1.92 at 24 against 2.40 at 24).

WHY CHOOSE THIS?

On CO2, a three-weight recursion beat the ARIMA model on the test months.

WHY NOT THIS? (TRADEOFFS)

One test of 24 months; whether the narrower intervals are honest needs many origins (section 5).

5. Rolling origins: accuracy and honesty of the intervals

DEFINITION — start here, no prior knowledge assumed

The same 97 origins as Day 149 (1990–1998): at each, every model is fitted on the months before it and forecasts 24 months ahead. The comparison now includes the intervals: a 95% interval should contain the truth about 95% of the time.

REAL-WORLD SCENARIO

A forecaster who says '90% sure' should be right about 9 times in 10; one who is right only 8 times in 10 is overconfident, however good their average guess.

ACTUAL OUTPUT

```
97 origins (186.5 s); all forecasts end by 1999-12
             MAE (ppm), months ahead       1       6      12      24
  Holt-Winters, multiplicative (AIC)   0.274   0.453   0.681   1.035
              Holt-Winters, additive   0.277   0.453   0.682   1.035
                  airline ARIMA, CSS   0.264   0.451   0.715   1.099
               same month last year   1.362   1.428   1.560   3.137
          95% coverage, HW additive   0.959   0.887   0.907   0.897   lowest 0.835
             95% coverage, airline   0.959   0.969   0.938   0.979   lowest 0.897
Holt-Winters (multiplicative) is more accurate than the airline model at 18 of 24 horizons (from 7 months ahead on); its
    intervals are too narrow, the airline model's are not
Actual output from this lesson's run.
```

What it means

Accuracy is nearly tied. The airline model is slightly better one month ahead (0.264 against 0.274); Holt-Winters is better from 7 months ahead on (18 of 24 horizons), reaching 1.035 against 1.099 at 24.

The intervals are not tied. The airline model's covered between 0.897 and 0.979; Holt-Winters' covered only 0.887 at 6 months, and as little as 0.835. They are too narrow. A likely reason, not tested here: the formula treats the fitted weights and the model as exactly right, and Holt-Winters' smaller in-sample error may understate how far forecasts drift.

WHY CHOOSE THIS?

Point forecasts as good as ARIMA's at a fraction of the conceptual cost.

WHY NOT THIS? (TRADEOFFS)

The textbook intervals were overconfident here; checking coverage on rolling origins is the only way to know.

6. Electrical-equipment orders

DEFINITION — start here, no prior knowledge assumed

On the log orders: rolling origins through the ordinary years 2001–2007, then 24-month tests through the crash and over the last two years. AIC chooses between damped and undamped Holt-Winters on each training stretch.

REAL-WORLD SCENARIO

A steady-demand estimate is fine until the market collapses; the question is how badly each method is caught out, and whether a simple rule would have done as well.

ACTUAL OUTPUT

```
rolling origins 2001-01 to 2007-01 (73), mean absolute % error by months ahead:
                               1       6      12
  Holt-Winters, additive   2.05%   4.74%   7.33%
      airline ARIMA, CSS   1.92%   4.39%   6.26%
    same month last year   5.31%   5.13%   4.83%
the crash: train up to 2007-12; AIC chose damped (alpha 0.328, beta 0.241, gamma 0.525, phi 0.916)
  MAE: Holt-Winters 18.81%, airline ARIMA 23.86%, same month last year 18.04%
the last 24 months: train up to 2014-05; AIC chose damped (alpha 0.668, beta 0.000, gamma 0.439, phi 0.980)
  MAE: Holt-Winters 4.65%, airline ARIMA 6.41%, same month last year 3.75%
in the ordinary period the airline model beats Holt-Winters at every horizon shown; through the crash Holt-Winters (damped,
as AIC chose) was much closer than the airline model, but neither beat last year's value in the crash or in the last two
    years
Actual output from this lesson's run.
```

What it means

In the ordinary years the airline ARIMA beat Holt-Winters at every horizon shown (1.92% against 2.05% one month ahead, 6.26% against 7.33% at 12), and same month last year beat both at 12 months (4.83%).

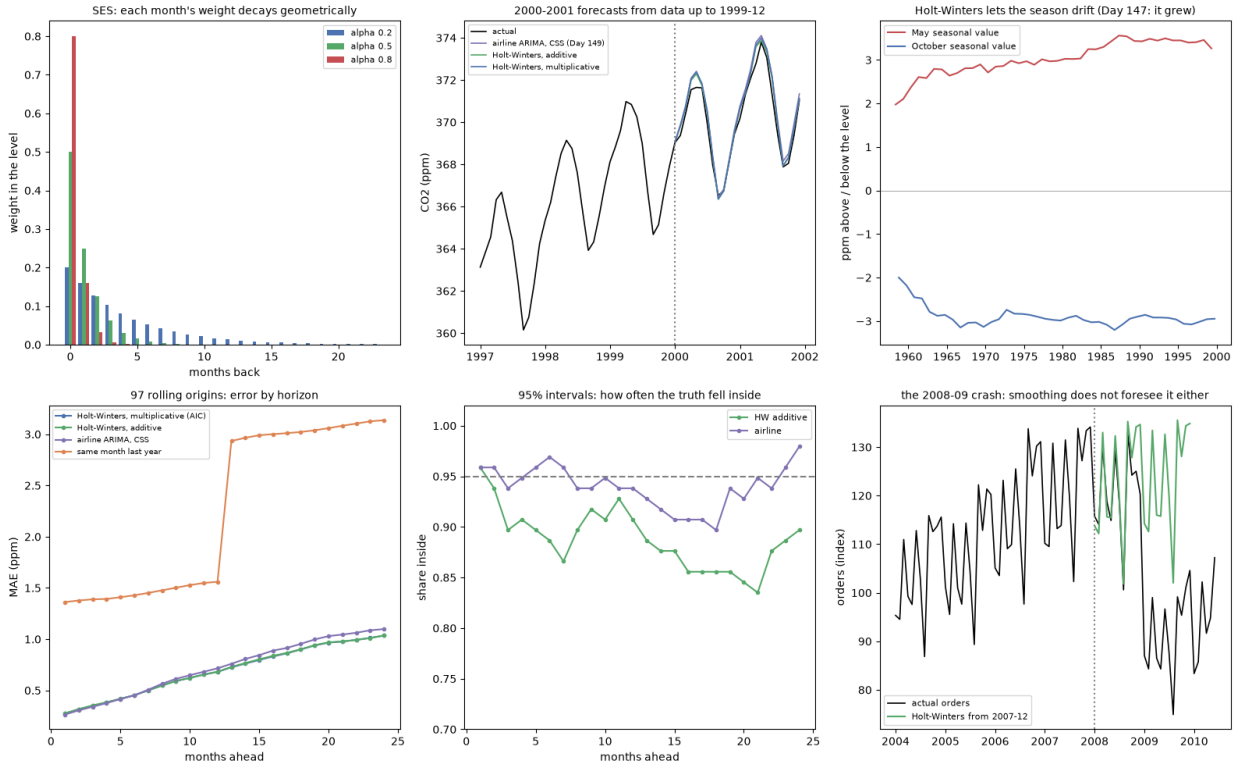
Through the crash, AIC chose a damped trend, and Holt-Winters (18.81%) was much closer than the airline model (23.86%), about as good as same month last year (18.04%). In the last two years last year's value won again (3.75% against 4.65% and 6.41%).

WHY CHOOSE THIS?

A damped trend keeps long-range forecasts from running away; in the crash AIC chose it, though whether it caused the better result was not tested.

WHY NOT THIS? (TRADEOFFS)

On the orders neither model beat the seasonal naive forecast a year ahead; a simple baseline must always be in the table.



Top: SES weights for three alphas; the 2000–2001 forecasts; Holt-Winters' May and October seasonal values over the training years. Bottom: rolling-origin error by horizon; interval coverage by horizon; the damped Holt-Winters forecast against the crash.

Interview preparation — technical questions

Q: How is simple exponential smoothing related to ARIMA?

A: It is ARIMA(0,1,1) with theta = alpha - 1: here the one-step errors agreed to 4.4e-15.

Q: How do you choose between additive and multiplicative seasonality?

A: Multiplicative when the seasonal swing grows with the level; here AIC on the training months chose multiplicative by 1.76, a narrow margin.

Q: Why fit from several starting points?

A: The least-squares surface has local minima: Holt from three starts stopped at 704.7799; a fourth start found 415.0547.

Q: How would you check a library's forecasts?

A: Against an independent implementation, and with a self-check: feed the forecasts back as data and confirm the one-step forecasts reproduce them (ours 1.1e-13, statsmodels 4.7e-02).

Q: Holt-Winters or ARIMA?

A: Measure both on rolling origins: on CO2 they tied on accuracy, but Holt-Winters' intervals covered as little as 0.835.

Interview preparation — theoretical questions

Q: Why 'exponential'?

A: The weight of a month k steps back is alpha (1 - alpha)^k, a geometric (exponential) decay.

Q: What does damping do?

A: It multiplies the trend by phi each step, so forecasts level off instead of extending a line forever.

Q: What do large and small smoothing weights mean?

A: Large: the estimate follows recent months (reactive, noisy); small: it changes slowly (steady, late to notice change).

Q: Why can a forecast interval be too narrow?

A: Its formula assumes the model and its fitted weights are exactly right; coverage on rolling origins showed 0.835 at worst instead of 0.95.

Key takeaway and what's next

KEY TAKEAWAY

Exponential smoothing forecasts from running estimates of level, trend and season, each nudged towards every new month. It is a short recursion, it contains SES = ARIMA(0,1,1) as a special case, and from scratch it matches statsmodels' fit, while a self-check exposed two forecast steps where statsmodels differs. On CO2, Holt-Winters matched or beat ARIMA on accuracy but its intervals were overconfident; on the orders, ARIMA was better in ordinary times and a damped Holt-Winters better in the crash, and last year's value remained hard to beat. Tomorrow turns forecasting into regression with features.

- SES equals ARIMA(0,1,1) with theta = alpha – 1 (4.4e-15).
- Same fit as statsmodels (sum of squares 60.7544); forecasts equal at 22 of 24 steps, and the self-check favours ours.
- CO2 2000–2001: Holt-Winters (AIC) 0.329 ppm against ARIMA 0.412.
- Rolling origins: tied on accuracy; Holt-Winters' 95% intervals covered as little as 0.835.
- Orders in the crash: damped Holt-Winters 18.81%, ARIMA 23.86%, last year's value 18.04%.

Suggested watch order, if starting from zero

- 1. Exponential Smoothing: Trends and Seasonality (Kevin Sheppard) — level, trend and seasonal smoothing explained, with the equations.
- 2. Smoothing 6: Winter's exponential smoothing (Galit Shmueli) — Holt-Winters (Winters') smoothing step by step, to compare with today's from-scratch version.

Day 151 preview

- **Feature engineering for time series.** Lags, rolling statistics and calendar features turn forecasting into ordinary regression. The lesson builds them from scratch and shows how easily future information leaks into such features, and what that does to a model's apparent accuracy.